

Gergely Bilkei-Gorzo

Specialist Embedded Systems



Personal Summary:

I am an automation engineer and PhD candidate at RWTH Aachen's Institute for Automotive Engineering, specializing in embedded systems development. I am passionate about advancing software and hardware solutions that drive efficiency, quality, and impactful results in the automotive industry.

Skills

Programming & Development:

- Proficient in C, Matlab/Simulink, and embedded systems (firmware, RTOS, bootloader, protocols, bus communication, circuit design)
- Experienced in Python and C++
- Skilled in containerization using POSIX functions
- Experience with container development using Docker
- Basic experience with application deployment via Kubernetes and CI/CD Pipeline

Hardware & Integration:

- Experienced in circuit and PCB layout design
- Extensive experience in hardware testing for series production designs
- Proven track record in system integration for prototype vehicles
- In-depth knowledge of Steer-by-Wire systems

Management:

- Experience in personnel and project management

Languages:

- Hungarian: Native
- German: Bilingual Proficiency
- English: Professional Working Proficiency

Contact

Gergely Bilkei-Gorzo
Vaalser Str. 150A, 52074 Aachen
Phone: +49 15122981347
Email: gergelybilkei@gmail.com

Work Experience

2018 - 2025 | Research Assistant/ PhD candidate

Institute for Automotive Engineering

- Designed, tested, and optimized hardware layouts for prototype and series production systems.
- Developed embedded software integrated into prototype vehicles.
- Managed customer relations and delivered technical lectures to industry partners.

2024 - 2025 | Specialist Embedded Systems Institute for Automotive Engineering

- Provided technical lectures for industry partners.
- Developed a container management system for automotive applications.

2025 - present | Expert Embedded Systems fka GmbH

- Developed software defined vehicle architecture for technical demonstration

Education

2016 - 2018 | Automation Engineering (RWTH Aachen)

- Master Thesis: Smart Power Distribution Unit for 48V automotive power net
- Worked as a student assistant on the design and control algorithm for a 400V to 48V DC/DC converter

2009 - 2015 | Mechanical Engineering, Specialization in Automotive (RWTH Aachen)